

MUHAMMAD HAZIQ HASBI BIN OSMAN

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PROFILE

Results-driven Electronic Engineering graduate with practical experience in machine maintenance, electronic component testing, and network system installation. Completed internships at Robert Bosch Semiconductor Manufacturing and SSIS E Solution. Skilled in soldering, troubleshooting, supporting spare part management, machine buy-off, and workpiece carrier repair and data tracking using Microsoft Excel. Strong understanding of industrial systems, equipment reliability, and electronic hardware functionality, with a commitment to quality and continuous improvement.

EDUCATION

Universiti Tun Hussein Onn Malaysia (UTHM) **Batu Pahat, Johor**
Bachelor's Degree with Honours in Electronic Engineering *(September 2021 – July 2025)*
CGPA: 3.18
Key Coursework: Embedded Systems, Microprocessor & Microcontroller, Analog & Digital Electronics, Computer Programming, Artificial Intelligence, Digital Signal Processing.

Politeknik Seberang Perai **Permatang Pauh, Penang**
Diploma in Electrical & Electronic Engineering *(Jun 2018 – August 2021)*
CGPA: 3.08
Key Coursework: Embedded Robotics, PLC, Semiconductor Devices, Electronic Circuits, Communication Systems, Equipment Repair.

WORK EXPERIENCED

Diploma Internship **Bukit Mertajam, Penang**
SSIS E Solution Sdn. Bhd. *(March 2021 – July 2021)*

- Assisted in the installation and configuration of fiber optic networks and server racks, ensuring compliance with network layout and safety standards.
- Performed Ethernet cable termination, routing, and labeling for LAN infrastructure to support high-speed data connectivity.
- Supported network equipment setup including switches, routers, and patch panels for new client installations.
- Performed signal testing, troubleshooting, and cable certification using network diagnostic tools to ensure optimal link performance.
- Maintained installation logs and updated network layout documentation for project delivery and audit compliance.

Degree Internship **Batu Kawan, Penang**
Robert Bosch Semiconductor Manufacturing Penang Sdn. Bhd. *(July 2024 – September 2024)*

- Managed and monitored spare part inventory for semiconductor equipment, ensuring 100% traceability and minimizing downtime due to part shortages.
- Developed an Excel-based spare parts tracking system with automated formulas and pivot analysis to improve data accuracy and retrieval efficiency by 30%.

- Supported machine buy-off activities, including functional verification, parameter validation, and documentation review before production release.
- Conducted testing and repair of workpiece carriers, identifying defective components and performing root cause analysis to restore functionality.
- Performed soldering and component replacement on PCB assemblies, ensuring compliance with ESD and IPC standards for rework quality.

ACADEMIC PROJECT

Prediction For Material Health Using Tree Based Machine Learning Model
Johor

Degree’s Final Year Project
(2025)

- Developed a machine learning model to predict maternal health risk levels (low, medium, high) based on clinical attributes such as blood pressure, blood sugar, and heart rate.

Energy Consumption Monitoring System with IOT
Johor

Degree’s Integrated Design Project
(2024)

- Designed and developed an IoT-based system to monitor and analyze real-time energy consumption for residential and small industrial environments, enabling remote tracking to optimize energy usage and reduce costs.

Autonomous Fire Fighter Robot
Penang

Diploma’s Final Year Project
(2021)

- Developed an autonomous fire-fighting robot to detect and extinguish small indoor fires without human intervention. The robot navigates autonomously, identifies fire sources using sensors, and activates a suppression mechanism for early fire control.

CERTIFICATE AND TRAINING

Integrated Design Project (IDP) Showcase - UTHM
Johor

Silver Award
(2024)

Training Matlab Calculus and Ordinary Differential Equation
Johor

(2024)

SULAM Project - FKEE
Johor

(2023)

Production Workshop Program – UTHM Multimedia Club
Johor

(2021)

Dean’s List Award
Johor

Sem 2 & 7
(2025)

SKILLS

Technical Skills: Microsoft Office, Microsoft Word, Microsoft Excel, Microsoft Teams, C ++ Programming, Python, MATLAB, Software Tools (Multisim, Arduino IDE, Proteus, Vivado, Xilinx SDK).

Practical Skills: Assembly Product, Preventive Maintenance, Buy-off Machine, Soldering, Basic Troubleshooting.

Language: Malay, English

REFERENCES

DR. NUR ILYANI BINTI RAMLI	ANBARASU BALASUBRAMANIAM
Supervisor Final Year Project UTHM	Supervisor Robert Bosch Semiconductor Manufacturing
013-4549494	016-4081259